

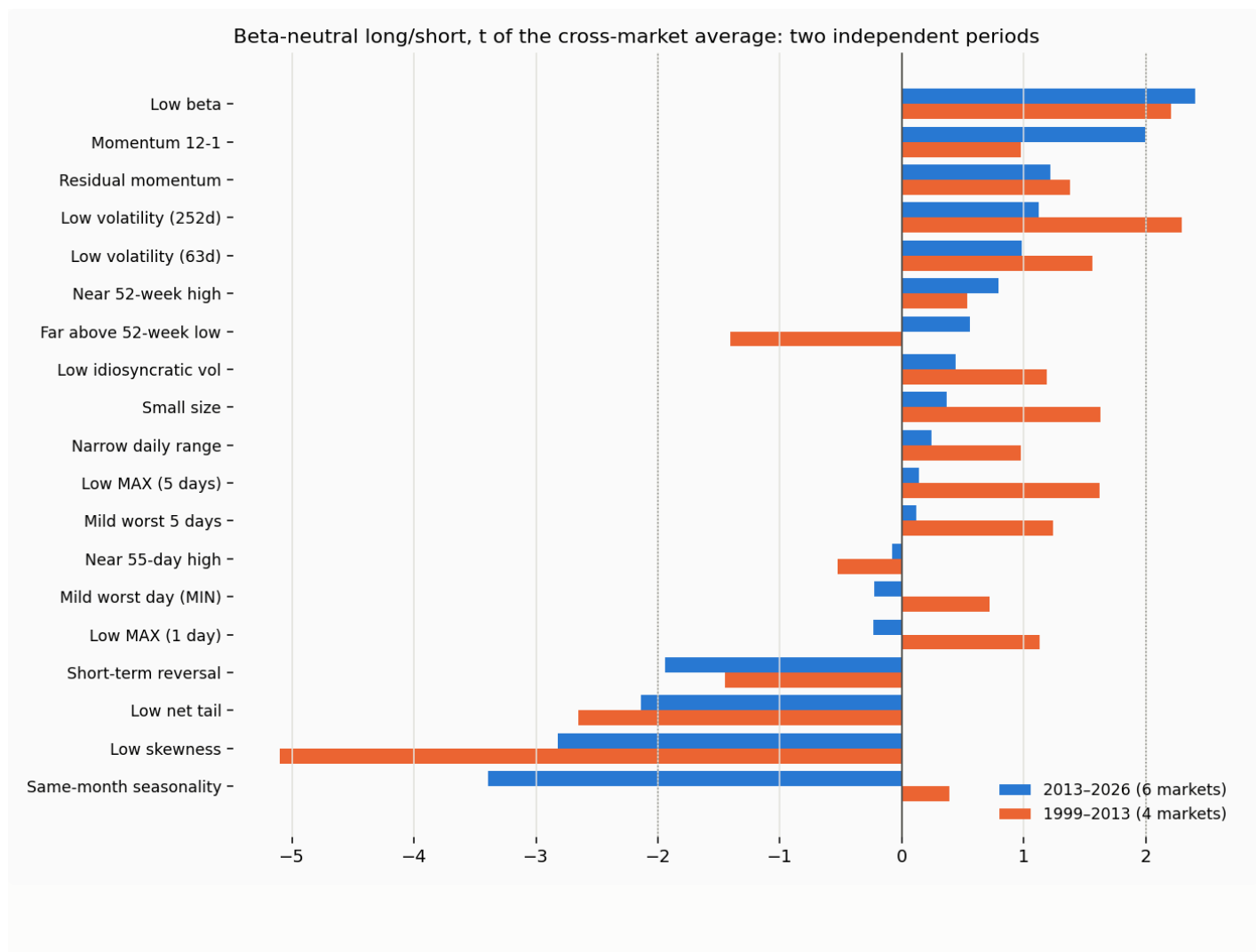
FACTSHEET FS15 · The earlier history

Do the conclusions hold in 1998–2013, with two bear markets they never saw?

Henrik Gjerner · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · ART data January 1998 – March 2013

Key facts	
Pre-registration	planning/PREREG_FS15.md, written and saved before any 1998–2013 return was computed; no deviations
Data	ART research database extract (Project2): daily total-return indices and traded value, as reported, including stocks that later delisted
Markets	US, EU (EU member states excl. UK, Poland, Greece), UK, Denmark; World = US + UK + EU in USD; no Scandinavia (Finland not identifiable in the extract)
Universes	top 494 US, 344 UK, 239 EU, 25 DK stocks by 63-day traded value each month (median eligible names); a liquidity universe, not an index
Engine	identical to FS13/FS13c/FS14: same 19 signals and directions, costs, borrow fees, financing, filter, pruning and weighting rules
Periods	signals Jan 1999 – Feb 2013; the FS14 walk-forward holds portfolios from Feb 2002 (36 months of history needed)

In one paragraph. The earlier period mostly agrees with the later one, and where it disagrees it is informative. **Low risk is confirmed:** beta-neutral low beta and low volatility earn t 2.2 and 2.3 across the four markets, positive in all four and in both halves. **Momentum is consistent but weaker** (12-1: t 1.0; flat through the 2008 crash, then hit hard in the spring 2009 rebound), and in the joint regression it again adds return (t 2.2, confirming FS13c). **Tail direction is confirmed as a loser** (low skewness t -5.1). **Seasonality reverses:** it paid before 2013 (joint slope t 4.3) and failed after, the pattern of a published anomaly that faded. The **FS14 dynamic selection fails again:** +0.6% a year over owning every book (t 0.2), positive before 2007 and negative after. The strongest result is the **fixed shortlist of low beta and 12-1 momentum**, chosen from 2013–2026 results and therefore genuinely out of sample here: Sharpe 0.81, +9.1% a year over owning every book (t 2.6), positive in all four markets, with a correlation of +0.03 to the market. It does not survive the programme-wide correction over 566 trials.



1. The price battery, 1999–2013

Signal	2013–26 beta-neutral t	1999–2013 beta-neutral, net / yr (t)	Markets positive	Sharpe 1999–2005 / 2006–13	1999–2013 raw L/S t	Verdict
Low volatility (252d)	+1.1	+15.9% (t +2.3)	4 / 4	+0.65 / +0.88	+0.9	confirmed
Low beta	+2.4	+19.5% (t +2.2)	4 / 4	+0.81 / +0.67	+0.9	confirmed
Small size	+0.4 (weak)	+6.1% (t +1.6)	4 / 4	+0.79 / +0.32	+1.6	consistent
Low MAX (5 days)	+0.1 (weak)	+8.6% (t +1.6)	4 / 4	+0.47 / +0.60	+0.8	consistent
Low volatility (63d)	+1.0 (weak)	+10.0% (t +1.6)	4 / 4	+0.42 / +0.58	+0.7	consistent
Residual momentum	+1.2	+5.8% (t +1.4)	4 / 4	+0.80 / +0.21	+1.5	consistent
Mild worst 5 days	+0.1 (weak)	+6.8% (t +1.2)	4 / 4	+0.27 / +0.54	+1.0	consistent
Low idiosyncratic vol	+0.4 (weak)	+5.2% (t +1.2)	4 / 4	+0.22 / +0.66	+0.9	consistent
Low MAX (1 day)	-0.2 (weak)	+5.7% (t +1.1)	4 / 4	+0.33 / +0.39	+0.4	reversed
Narrow daily range	+0.2 (weak)	+5.2% (t +1.0)	4 / 4	+0.18 / +0.46	+0.6	consistent

Signal	2013–26 beta-neutral t	1999–2013 beta-neutral, net / yr (t)	Markets positive	Sharpe 1999–2005 / 2006–13	1999–2013 raw L/S t	Verdict
Momentum 12-1	+2.0	+5.9% (t +1.0)	4 / 4	+0.23 / +0.34	+1.0	consistent
Mild worst day (MIN)	-0.2 (weak)	+3.6% (t +0.7)	3 / 4	+0.11 / +0.35	+0.7	reversed
Near 52-week high	+0.8 (weak)	+3.1% (t +0.5)	3 / 4	-0.01 / +0.34	+0.8	consistent
Same-month seasonality	-3.4	+1.0% (t +0.4)	2 / 4	-0.15 / +0.44	+0.4	reversed
Near 55-day high	-0.1 (weak)	-2.7% (t -0.5)	1 / 4	-0.31 / +0.04	+0.3	consistent
Far above 52-week low	+0.6 (weak)	-7.7% (t -1.4)	0 / 4	-0.65 / -0.11	+0.1	reversed
Short-term reversal	-1.9	-6.7% (t -1.5)	0 / 4	-0.24 / -0.80	-2.6	consistent
Low net tail	-2.1	-7.6% (t -2.7)	0 / 4	-0.62 / -1.12	-2.9	confirmed
Low skewness	-2.8	-10.8% (t -5.1)	0 / 4	-1.58 / -1.48	-4.2	confirmed

Verdict as pre-registered: confirmed = same sign as 2013–26 and $|t| > 2$ in 1999–2013; consistent = same sign, $|t| \leq 2$; reversed = opposite sign. "(weak)" marks a 2013–26 result with $|t| < 1$, which barely had a sign to confirm. Averages over US, EU, UK and DK; World is left out because it overlaps.

Confirmed: Low net tail, Low skewness, Low volatility (252d), Low beta. **Reversed:** Low MAX (1 day), Mild worst day (MIN), Same-month seasonality, Far above 52-week low. Everything else keeps its sign with a t below 2.

2. All signals together (Fama-MacBeth), 1999–2013

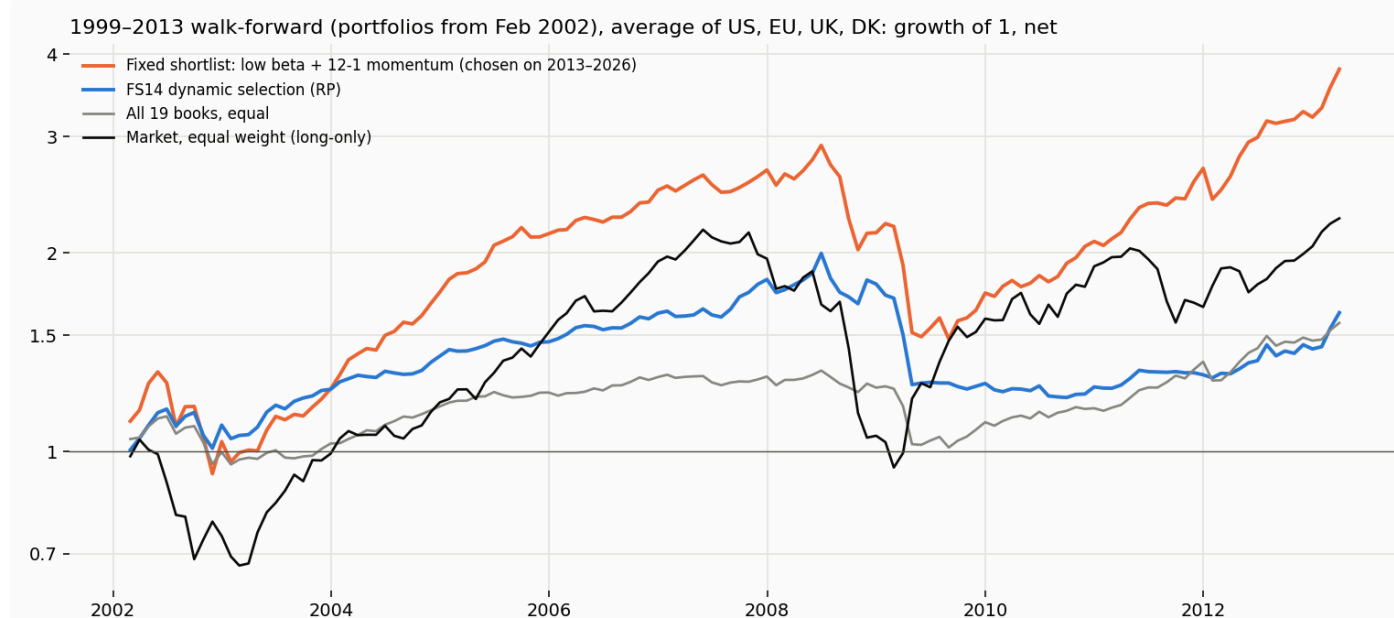
Signal	2013–26 joint slope t	1999–2013 joint slope t	Markets positive	Verdict
Low beta	-0.9	+0.2	1 / 3	reversed
Low volatility (252d)	-1.2	+1.2	3 / 3	reversed
Low MAX (1 day)	+0.7	+3.2	3 / 3	confirmed
Momentum 12-1	+2.7	+2.2	3 / 3	confirmed
Residual momentum	-0.4	+0.4	2 / 3	reversed
Near 52-week high	+0.2	-1.7	1 / 3	reversed
Short-term reversal	-0.4	+1.2	3 / 3	reversed
Same-month seasonality	-0.6	+4.3	3 / 3	reversed
Low skewness	+0.7	-3.8	0 / 3	reversed
Small size	+1.4	+3.4	3 / 3	confirmed

12-1 momentum keeps a positive joint slope in both periods, the one price signal that does. The earlier period also rewards signals that are weak or negative after 2013: seasonality, low MAX and small size.

3. The FS14 procedure, 1999–2013

Market	RP (primary)	W12- X	All books equal	Fixed shortlist	Market	Books held (median)	Months in cash
US	-0.01	+0.20	-0.00	+0.27	+0.37	2	17
EU	+0.22	+0.25	+0.36	+0.67	+0.27	3	12
UK	+0.67	+0.93	+0.69	+0.75	+0.52	3	16
DK	+0.32	+0.71	+0.69	+0.93	+0.66	3	7
World	+0.17	+0.42	+0.40	+0.74	+0.43	2	19
4-market average	+0.46	+0.79	+0.54	+0.81	+0.48	—	—

Sharpe ratios, net, Feb 2002 – Mar 2013. Fixed shortlist = low beta and 12-1 momentum, beta-neutral, equal weights.



	2016–2026 (FS14)	2002–2013 (FS15)
Primary: RP minus ALL-EQ	+2.2% (t +1.4)	+0.6% (t +0.2)
Best weighting rule, Sharpe	0.45	0.79 (W12-X)
RP, Sharpe	0.34	0.46
All books equal, Sharpe	0.03	0.54
Fixed shortlist, Sharpe	0.50 (hindsight)	0.81 (out of sample)

Primary test: RP – ALL-EQ = +0.6% a year, t 0.25 (+5.3% up to 2007-09, -4.3% after): **not passed**, for the second time.

The shortlist: +9.1% a year over owning every book (t 2.62), positive in 4 of 4 markets and in both halves (+12.9% / +5.1%). Because the shortlist was chosen from 2013–2026 results, this is a clean out-of-sample test of that choice.

4. Bear markets

Book (4-market average)	Dot-com bust (Apr 2000 – Sep 2002)	Financial crisis (Nov 2007 – Feb 2009)	Momentum crash (Mar – May 2009)	Euro crisis (May – Sep 2011)
Market (equal weight)	-42%	-56%	+34%	-23%
Low beta, beta-neutral	+124%	-29%	-21%	+11%
12-1 momentum, beta-neutral	+1%	+2%	-42%	+4%
Shortlist (low beta + momentum)	+60%	-14%	-32%	+8%
Low skewness, beta-neutral	-31%	-5%	+1%	-7%

Cumulative return over each episode. The beta-neutral books are long/short and market-neutral by construction.

Low beta was the great protector in the dot-com bust and paid in the euro crisis, but lost in the financial crisis, when its leveraged low-beta leg fell with everything else. Momentum held flat through 2008 and then lost heavily in the spring 2009 rebound, when its short leg of beaten-down stocks rallied hardest (Daniel & Moskowitz 2016). The two legs fail at different times, which is why the pair beats either alone.

5. What this adds to the series

1. **Low beta and momentum keep their sign across 27 years and two independent samples** (low beta significant in both, momentum only in the joint regression before 2013), and together they beat both the dynamic selection and owning everything.
2. **The dynamic selection process (FS14) failed its pre-registered test twice.** A rules-based filter on trailing Sharpe ratios chases what just worked and lags turning points.
3. **Seasonality and low MAX worked before 2013 and not after** (small size weakened), the decay McLean & Pontiff (2016) and JKP describe after publication and wider use.
4. **Tail direction (low skewness) loses in both periods:** in large caps the stocks with the most positive skew kept earning more, not less.

6. Caveats

- **Liquidity universe.** Without membership dates, the universe is the most traded stocks each month, larger than the 2013–26 index universes in the UK and US and not identical to them.
- **As-reported prices** from one research database; bad ticks are filtered (daily moves above 100%, and +50%/–33% one-day reversals), but data quality is below today's vendors.
- **Currency.** Traded value is converted to USD with Fed H.10 rates for ranking; pre-1999 euro-area values use the first euro rate.

- **Overlap.** The ART extract ends in March 2013 and the 2013–26 sample starts in February 2013: two months overlap.
- **Multiple testing.** FS15 adds 157 trials to the ledger (now 566).

7. References

- Daniel, K. & Moskowitz, T. J. (2016). Momentum crashes. *Journal of Financial Economics* 122(2), 221–247.
- McLean, R. D. & Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance* 71(1), 5–32.
- Frazzini, A. & Pedersen, L. H. (2014). Betting against beta. *Journal of Financial Economics* 111(1), 1–25.
- Heston, S. L. & Sadka, R. (2008). Seasonality in the cross-section of stock returns. *Journal of Financial Economics* 87(2), 418–445.
- Asness, C. S., Moskowitz, T. J. & Pedersen, L. H. (2013). Value and momentum everywhere. *Journal of Finance* 68(3), 929–985.
- Jensen, T. I., Kelly, B. & Pedersen, L. H. (2023). Is there a replication crisis in finance? *Journal of Finance* 78(5), 2465–2518.

8. Reproduce

`planning/PREREG_FS15.md` → `code/data.py` (`load_art`, `universes`) → `code/signals.py`, `signals2.py` (`regions US_A, EU_A, UK_A, DK_A, WD_A`) → `code/art_run.py` (`battery`, Fama-MacBeth, FS14 procedure → `results/art_*.json`) → `code/build_fs15.py`. The ART extract is private and not published.